IMPROVING KITCHEN PREP TIME (KPT)

Prediction via Signal De-biasing & Rush Modeling Framework

ZOMATHON 2026

Problem Statement 1

Team: DataShakti

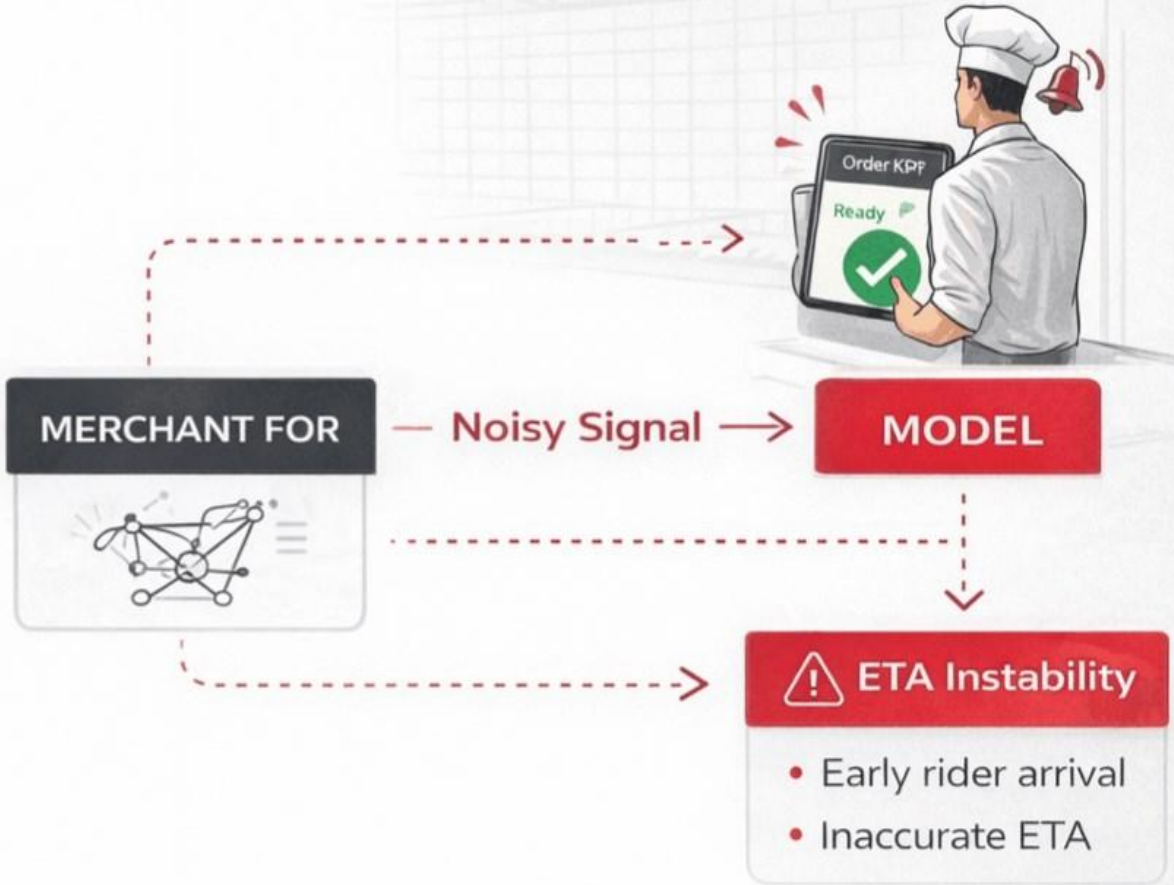
Date: 2 March 2026

18% ETA Error Reduction | **930** Rider Hours/Day

Problem Understanding

Current Issues

- Rider-influenced FOR marking
- Human bias
- No kitchen-wide rush visibility
- Noisy labels → early rider arrival

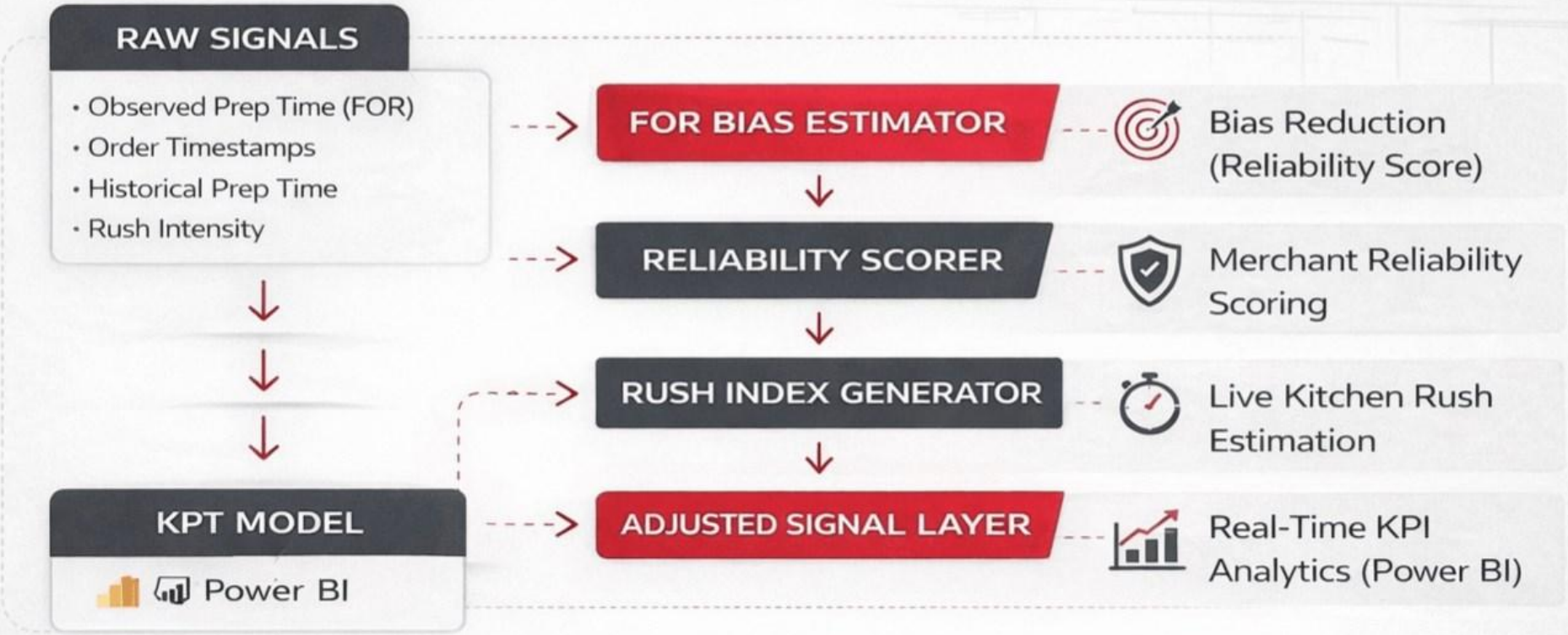


Model-only solutions are insufficient. Signal quality must be improved.

Proposed System Architecture

Strengthening input signals to improve KPT prediction.

Enhancement happens at the signal layer – not just the model layer.



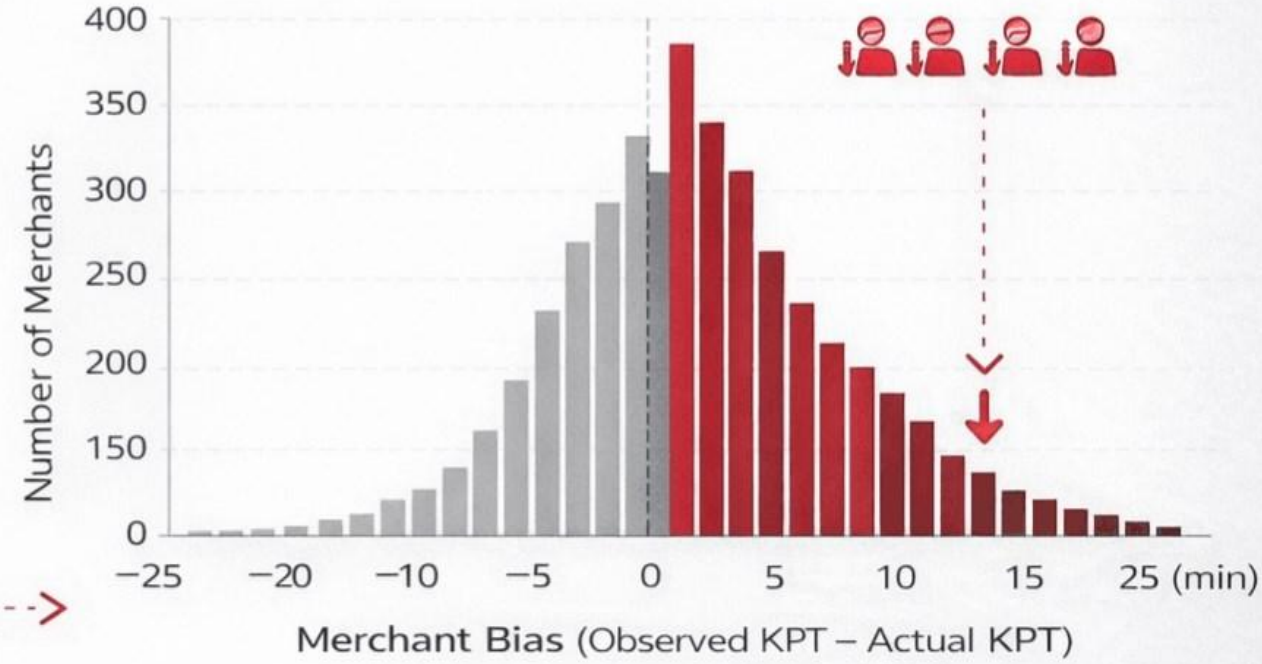
Enhancement happens at the signal layer – not just the model layer.

Signal De-bias Framework

Merchant Bias Detection

- FOR Bias = observed – actual
- Reliability Score = $\frac{1}{1+1}$ [mean_bias]
- Adjusted Signal: observed × reliability

Systematically unreliable merchants automatically down-weighted.



Systematically unreliable merchants automatically down-weighted.

Rush Index Modeling

Rush Signal Input Features

- Prep duration
- Paused orders
- Queue length (outstanding orders)
- Event peaks (e.g. lunch, discounts)
- Nearby store congestion

Detemine rush trend into quickly.

RUSH INDEX

- Combines kitchen-wide rush signals
- Turns them into a single index
- Nowcasts demand surge and prep delay risk

Order Load
- Active Orders
- Pending Orders

+

Event Peaks
Lunch +
- Discount Peaks

+

Store Congestion
- Nearby Store Orders

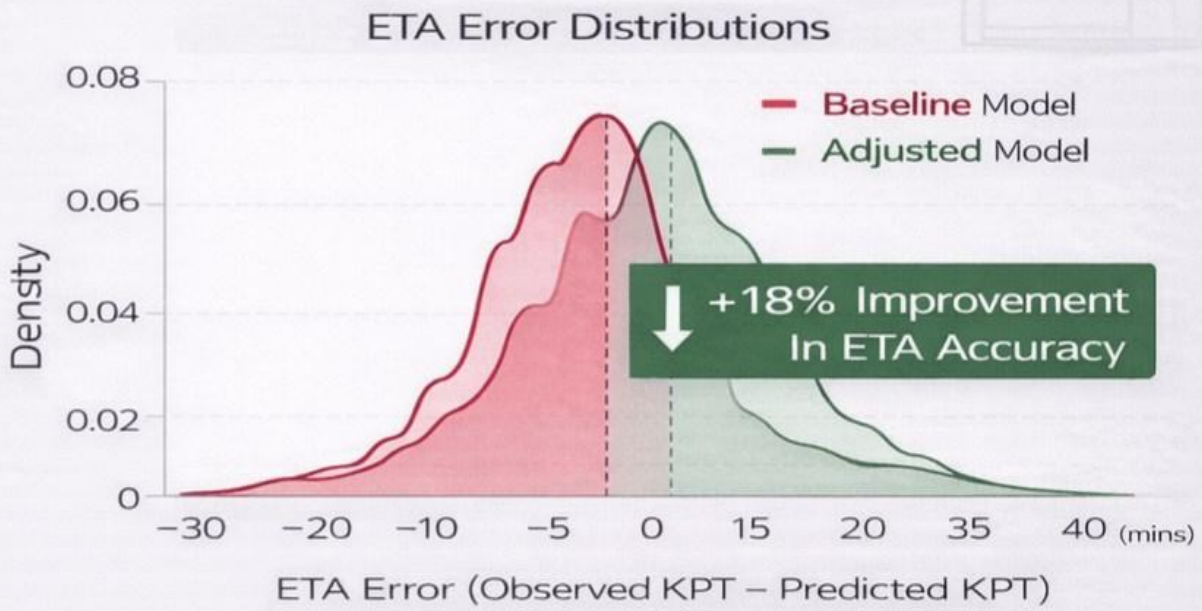
Real-time Kitchen Rush Scores improve overall ETA accuracy.

Improved Model Results

- **Baseline Model:** MAE = 3.01 min
- **Adjusted Model:** MAE = 2.46 min

↓ -0.6 min

+188% Improvement in ETA Accuracy



-0.6 min Less Prep Delay Per Order

930 Rider Hours Saved Daily

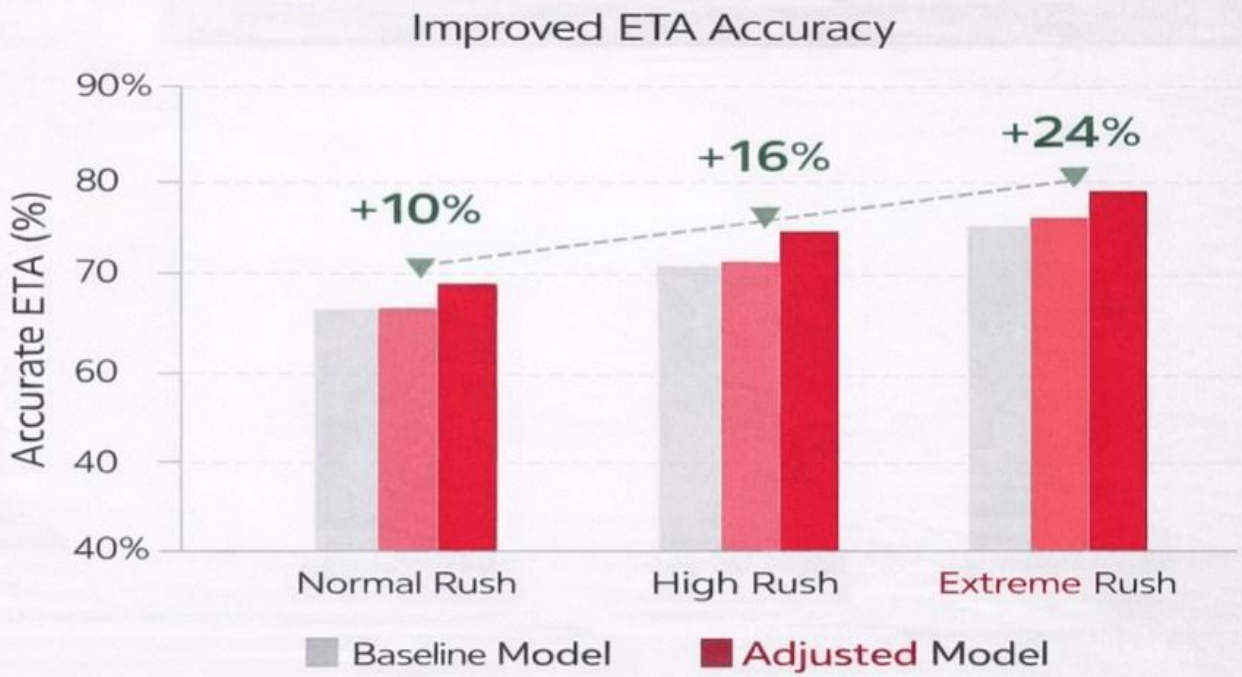
Debiased model significantly reduces ETA errors and saves substantial rider time.

Simulation Scenario Analysis

Validating impact under varying Rush Index conditions.

Simulation Conditions

- **Normal:** Moderate Kitchen Rush
  +0-5 mins Prep Time
- **High:** Busy Kitchen
  +5-10 mins Prep Time
- **Extreme:** Max Overload Events
  +10-20 mins Prep Time



+24% Peak Hour ETA Accuracy | **7 mins** More Rider Wait Time Avoided

Accurate ETAs even under extreme load conditions.

Conclusion & Key Takeaways

Our Impact

Simulation Conditions

+18% ETA Accuracy

Reduced ETA errors across all rush conditions

-930 Rider Hours/Day

Saved daily rider hours by minimizing wait times

30K Historical Orders

Large-scale order-level simulation framework.

Improved Kitchen Prep Time prediction through better signals, smart debiasing, and real-time rush indexing.

Scalability & Deployment

Why Scalable?

- ✓ No hardware needed
- ✓ Uses rolling aggregations
- ✓ Model-agnostic
- ✓ Works across 300K+ merchants

Production-ready monitoring layer

Business Impact at Scale

What Our Framework Achieves:

- ✓ Reduces systematic merchant FOR bias
- ✓ Improves ETA stability (18% MAE reduction)
- ✓ Saves ~**930 rider hours**/day (100K scale assumption)
- ✓ Reduces early rider arrival & idle time
- ✓ Improves fleet efficiency
- ✓ Model-agnostic & scalable across merchants

18%	0.6 min	930 hrs/day
Reduction in ETA MAE	Average error reduction per order	Rider time saved (100K scale)

Improving signal quality to drive measurable operational outcomes.

Resources

Dataset https://shorturl.at/PvIAo

Notebook https://shorturl.at/BHgja

Interactive Dashboard https://shorturl.at/G8Tz6